

Topological Memory - Dependence

Introduction Today:

Dependence.

Local Stochastic Noise.

$$\begin{aligned}
 & \text{Dependence Type I [h] [scale=0.7]} \\
 & \text{node (u1) at (3,3) [draw,shape=rectangle]u}_1; \text{ node (x) at (0,6) [draw,shape=circle]x; node (y) at (0,0) [draw,shape=circle]y; } \\
 & \Pr[(x, y) \in e | u_1 \text{ is satisfied}] \\
 = & \Pr[x \in e | u_1 \text{ is satisfied}] \Pr[y \in e | u_1 \text{ is satisfied}] \\
 \leq & \Pr[x \in e] \Pr[y \in e] \\
 & \text{Dependence Type II [h] [scale=0.7]} \\
 & \text{node (u1) at (3,1) [draw,shape=rectangle]u}_1; \text{ node(u2) at (3,4) [draw,shape=rectangle]u}_2; \text{ node (x) at (0,6) [draw,shape=circle]x; node (y) at (0,0) [draw,shape=circle]y; } \\
 & \Pr[(x, y) \in e | z \text{ is not flipped}] \\
 = & \Pr[x \in e | z \text{ is not flipped}] \Pr[y \in e | z \text{ is not flipped}] \\
 \leq & \Pr[x \in e] \Pr[y \in e]
 \end{aligned}$$

Definition Consider two (qu)bits x, y . We say that z is a **linked node** between x and y if z is either x or y .

Remark Notice that y might be a **linked node** between x and y .

Definition Consider an application of a noise channel. We say that z is a **clean linked node** if:

z is a bit and $z \notin f$ (not flipped).

z is a check and z is satisfied.

$$\begin{aligned}
 & \Pr[x_1, x_2, \dots, x_k, x_{k+1}, \dots, x_l \in e | z_1, z_2, \dots, z_m] \\
 = & \prod_i \Pr[x_i \in e | z_1, z_2, \dots, z_m] \\
 = & \prod_i \Pr[x_i \in e | z_i \in \Gamma(x_i)] \\
 \leq & \prod_i^k \Pr[x_i \in e] \prod_{i=k+1}^l \Pr[x_i \in e | z_i \in \Gamma(x_i)] \\
 \leq & M(p)^k
 \end{aligned}$$

Local Decoder

definition We say that D is a local decoder if when applied against $2p(1-p)$ -depolarized channel:

There exists function $M : (0, 1) \rightarrow (0, 1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p^{\frac{100}{99}}$.

There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n, k) -Toric, then both the swift-rule and the flip upon majority rule are local decoders.

Definition Consider the moment after the application of the depolarizing channel, and right before the application of the decoder.

For a subset of qubits S , we say that a qubit $x \in S$ is **dependent** if there exists $y \in S$ such:

Either x and y share a check that is not satisfied. (Type I)

Or, x and y have checks which share a bit that was flipped by the application of the channel. (Type II)